Quantum Mechanics as a Forced Structural Consequence of Event Density Primitives

Allen Proxmire

April 2026

Abstract

The non-relativistic quantum mechanics of a single particle is shown to arise as a structural consequence of the Event Density (ED) primitive stack. A single mathematical object — the participation measure $P_K(x,t) = \sqrt{b_K(x,t)}\, e^{i\pi_K(x,t)}$, a complex-valued distribution over channels K and positions x — fuses two ED primitives (bandwidth and tension polarity) into the carrier of quantum-mechanical content. In a well-defined thin-participation limit, the coherent sum $\Psi = \sum_K P_K$ satisfies the standard linear Schrödinger equation. Environmental phase-randomization at commitment events, together with an individuation threshold, forces the Born rule in its squared-amplitude form as a definitional consequence of $b_K = |P_K|^2$. Non-factorizable joint participation measures on bipartite systems produce Bell-inequality-violating correlations that saturate the Tsirelson bound $2\sqrt{2}$. The Fourier-conjugate decomposition of the adjacency-band bandwidth budget, combined with the Fourier-uncertainty theorem, yields the Heisenberg uncertainty relation $\Delta x \cdot \Delta p \geq \hbar/2$. A single structural commitment — the identification of bandwidth with squared amplitude, $b_K = |P_K|^2$ — threads through five otherwise-independent appearances of the number two in quantum mechanics: the Born rule, the Schrödinger kinetic term, the Madelung amplitude-phase decomposition, the sublinear bandwidth-composition rule, and the Fourier-uncertainty bound. The program derives the form of quantum-mechanical equations; the numerical values of the dimensional constants $\hbar$ and m are inherited from the Event Density Dimensional Atlas rather than derived from primitives, and the primitive-level derivation of particle masses is flagged as a frontier research problem. No physical postulate beyond the participation-measure structure, the four-band bandwidth decomposition, and the commitment dynamics is required. The result is a structural reduction of the four independent postulates of standard quantum mechanics to consequences of a single primitive-level framework.

---

1. Introduction

Standard non-relativistic quantum mechanics rests on four logically independent postulates: states are complex vectors in a Hilbert space; state evolution is governed by the Schrödinger equation; measurement outcomes occur with probabilities given by the Born rule; physical observables are represented by self-adjoint operators whose commutation relations produce the uncertainty relations. Each postulate has been verified to extraordinary empirical precision, and each plays a well-defined role in the predictive machinery of the theory. At the foundational level, however, no derivation is

known within conventional formulations that reduces any of the four postulates to consequences of the others, and several foundational programs have attempted to supply such a reduction from various directions. The Many-Worlds framework attempts to derive the Born rule from decision-theoretic rationality axioms applied to branching dynamics [wallace2012]. Gleason's theorem derives the Born rule from assumptions of non-contextuality and additivity on a Hilbert-space projector lattice, but the Hilbert-space structure itself is assumed rather than derived [gleason1957]. Bohmian mechanics reproduces the predictions of non-relativistic quantum mechanics using particle positions and a non-local pilot wave [bohm1952]. Relational and QBist programs reinterpret the postulates at the interpretive level without modifying the formal content [rovelli1996, fuchs2002]. None of these programs reduces all four postulates to consequences of a single underlying structural framework.

This paper presents such a reduction within the Event Density (ED) program. The program starts from a discrete ontology of events, chains, participation relations, and commitment dynamics. A distinguished object — the participation measure — is identified as the carrier of quantum-mechanical content within this ontology. Under a clearly specified thin-participation limit, the five-step derivation presented in sections 3 through 7 produces each of the four quantum-mechanical postulates as a structural consequence of the participation-measure framework.

The argument is organized as follows. Section 2 describes the ED primitive stack used in the derivation. Section 3 defines the participation measure and establishes its structural properties. Section 4 derives the Schrödinger equation as the thin-participation limit of a linear first-order complex evolution equation for the participation measure. Section 5 derives the Born rule by showing that environmental phase-randomization at commitment events, combined with an individuation threshold, forces channel-selection probabilities to take the bandwidth-fraction form, whose squared-amplitude structure is definitional. Section 6 derives Bell-inequality-violating correlations from non-factorizable joint participation measures and obtains the Tsirelson bound from the sesquilinear inner product of the participation manifold. Section 7 derives the Heisenberg uncertainty relations from the Fourier-conjugate partition of the adjacency-band bandwidth budget. Section 8 presents a structural observation — the exponent-two thread — that connects the Born rule, the Schrödinger kinetic term, the Madelung decomposition, the sublinear bandwidth-composition rule, and the Fourier-uncertainty bound to a single definitional commitment. Section 9 distinguishes structural content, which is derived from primitives, from dimensional anchoring, which is inherited through the ED Dimensional Atlas. Section 10 catalogues limitations and open research frontiers, and section 11 concludes.

2. The Event Density Framework

This section summarizes the ED primitive stack used in the derivation. A complete account of the primitive stack is available in the companion Event Density literature; the presentation here is restricted to the structural content directly used in the derivations that follow.

The ED ontology is a graph-theoretic structure whose vertices are micro-events and whose edges carry participation relations. A chain is a sequence of micro-events held together by a consistent update rule; the rule persists across the sequence and constitutes the chain's identity. Participation is the relational substrate connecting micro-events; two events participate when each integrates the other's becoming into its own. Participation is not a force, field, or exchange of information at the primitive level, but the relational fabric on which forces, fields, and information-bearing structures are coarse-grained.

The scalar magnitude of a participation relation is called the participation bandwidth. For a chain

C at position x and time t, the total bandwidth budget $b_{\text{total}}(C)$ decomposes into four distinguished bands:

$$b_K = b_K^{\text{int}} + b_K^{\text{adj}} + b_K^{\text{env}} + b_K^{\text{com}}, \tag{1}$$

where b_K^{int} denotes internal rule-bandwidth sustaining the chain's own update rule, b_K^{adj} denotes adjacency bandwidth with the chain's immediate participation-adjacent neighborhood, b_K^{env} denotes environmental bandwidth shared with the broader bath, and b_K^{com} denotes commitment-reserve bandwidth available for commitment events. The index K ranges over the channels available to the chain at the position and time of interest.

A channel is a stable bandwidth-preserving sub-structure of the participation graph that supports a chain's update rule repeatedly without rule-incompatibility. Channels are discrete at the graph level and become a continuous index in the coarse-grained thick-regime description. The multiplicity M of a chain at a position is the count of distinct channels available for continuation. A commitment event is the discrete selection of a single channel by a chain, together with the addition of a new micro-event along the selected channel's next-vertex position. The commitment event is local at the participation site; its trigger condition is the rapid growth of environmental bandwidth past an individuation threshold. Tension polarity is the phase relation between the chain's update rule and the local flow direction; it takes values in S^1 and provides the phase content that the participation bandwidth, a real non-negative scalar, does not carry on its own.

The thin-participation regime is the limit in which the effective multiplicity satisfies $M_{\text{eff}} \gg 1$, the environmental band satisfies $b_K^{\text{env}} \to 0$, and the commitment rate satisfies $\Gamma_{\text{commit}} \to 0$. In this regime the chain is distributed coherently across many channels with no environmental decoherence and no commitment events, constituting the primitive-level analogue of an isolated quantum system.

3. The Participation Measure

The first step of the derivation is the definition of a single mathematical object that carries both the amplitude content of the participation bandwidth and the phase content of the tension polarity.

3.1 Definition

The participation measure is a complex-valued distribution over channels and positions, defined by

$$P_K(x,t) = \sqrt{b_K(x,t)}\, e^{i\pi(K,x,t)}, \tag{2}$$

where $b_K \geq 0$ is the bandwidth in channel K at position x and time t, and $\pi(K,x,t) \in [0, 2\pi)$ is the tension-polarity phase. By construction,

$$b_K(x,t) = |P_K(x,t)|^2, \tag{3}$$

so the participation bandwidth is the squared amplitude of the participation measure. The event density, a scalar field that counts micro-events per region, is recovered as the sum over channels:

$$\rho(x,t) = \sum_K |P_K(x,t)|^2 = \sum_K b_K(x,t). \tag{4}$$

The coherent sum over channels defines a complex scalar field

$$\Psi(x,t) = \sum_K P_K(x,t), \tag{5}$$

which plays the role of the quantum-mechanical wavefunction in the thin-participation limit.

3.2 Structural constraints

The participation measure inherits several structural constraints from the primitive stack. The measure must carry both amplitude and phase, because the Born rule requires bandwidth weighting and interference requires phase differences. The measure must support linear superposition of the form $\alpha P^A + \beta P^B$ for complex scalars α, β, because coherent recombination of channels at interferometric junctions preserves the superposed structure. The measure must admit a sesquilinear inner product of the form

$$\langle P \,|\, Q \rangle = \sum_K \int dx \, P_K^*(x) \, Q_K(x), \tag{6}$$

which yields a real non-negative norm equal to the total participation bandwidth. The measure must be decomposable across the four bands of equation~(1). Joint participation measures on multipartite systems must admit non-factorizable configurations, corresponding to entangled states. Finally, the measure must be compatible with the commitment dynamics, in which the bandwidth-fraction $|P_{K^*}|^2 / \sum_K |P_K|^2$ provides the selection probability at a commitment event.

3.3 Uniqueness of the algebraic structure

The participation measure must carry amplitude and phase, support linear superposition with complex coefficients, and admit a sesquilinear inner product. The minimum algebra satisfying these requirements is the complex numbers $\mathbb{C}$. Real scalars alone cannot support phase rotations; the natural pair (b, π) under custom operations is isomorphic to $\mathbb{C}$ via equation~(2). Quaternions support all the required operations but introduce non-commutative multiplication, which conflicts with the symmetric structure of shared participation between endpoints of an entangled pair. Higher-dimensional algebras are over-structured relative to the primitive-level requirements. The complex numbers constitute the unique minimum algebra for the participation measure.

The square-root structure of equation~(2) is forced by the Madelung-class identification of the event density with the squared modulus of the wavefunction in the quantum regime. Under this identification, the bandwidth b_K must equal $|P_K|^2$ rather than $|P_K|$, and the polar decomposition of a complex number fixes the amplitude of P_K as $\sqrt{b_K}$.

4. The Schrödinger Equation

The participation measure evolves in time according to a linear first-order complex equation whose specific form is derivable from the primitive stack together with the structural properties established in section 3.

4.1 Derivation of the evolution equation

The evolution of the participation measure is linear. Linearity follows from the requirement that superposition be preserved under evolution, which in turn follows from the multi-channel thin-regime structure and the compatibility of the commitment dynamics with the Born rule. A non-linear evolution would violate the preservation of superposition, produce Born-rule inconsistency at successive commitment events, or permit superluminal signaling in entangled systems; each of these consequences is structurally excluded by the primitive-level commitments.

The evolution is first-order in time. The chain's update rule applies one step at a time; the rule's next application depends only on the current state of the chain, not on past history. This is the primitive-level content of a Markov-type dynamics, and it rules out higher-order time derivatives.

The most general linear first-order evolution compatible with these constraints is

$$i\hbar\,\partial_t P_K = H_K\,P_K + \sum_{K'} V_{KK'}\,P_{K'}, \tag{7}$$

where H_K is a channel-diagonal term and $V_{KK'}$ represents off-diagonal inter-channel coupling. The factor $i\hbar$ on the left-hand side is forced by the following argument. Total bandwidth of an isolated chain is conserved in the thin-participation regime, so

$$\frac{d}{dt}\sum_K |P_K|^2 = 0. \tag{8}$$

Substituting equation~(7) and requiring this vanishing for arbitrary P yields the anti-Hermitian condition $L_{KK'} + L^*_{K'K} = 0$ on the evolution kernel L. Any anti-Hermitian operator can be written as $L = -iH/\hbar$ with H Hermitian; the prefactor $i\hbar$ then appears on the time-derivative side, yielding equation~(7).

The numerical value of $\hbar$ is inherited from the ED Dimensional Atlas through the Madelung-class identification of the quantum regime with the free-particle Schrödinger equation, whose diffusion coefficient $D = \hbar/(2m)$ is a mathematical identity under the Madelung transformation. The structural form of equation~(7) is derived from primitives; the specific numerical value of $\hbar$ is inherited.

4.2 Thin-participation limit

In the thin-participation limit the channel index K becomes continuous. Under translation invariance of the participation graph, the natural continuous index is the momentum k; Stone's theorem on unitary one-parameter groups [stone1930] identifies the generator of translations as the momentum operator $\hat{P} = -i\hbar\partial_x$. In the momentum representation, the diagonal term of equation~(7) takes the form $H_k = \hbar^2 k^2/(2m) + V$, where the quadratic dependence on k is forced by rotation invariance, analyticity of the dispersion relation at $k = 0$, and the non-relativistic regime; linear-in-k and higher-order-in-k terms are excluded. The coefficient $1/(2m)$ follows from classical-correspondence (Ehrenfest) constraints, with the mass parameter m inherited empirically at the dimensional-anchoring level.

Summing equation~(7) over channels and identifying the result with $\Psi = \sum_K P_K$ yields

$$i\hbar\,\partial_t\Psi(x,t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right]\Psi(x,t), \tag{9}$$

the standard linear Schrödinger equation for a particle of mass m in the potential $V(x)$.

4.3 Equivalence to the Madelung form

Polar decomposition of the wavefunction as $\Psi = \sqrt{\rho}\,e^{iS/\hbar}$, substituted into equation~(9) and split into real and imaginary parts, yields the continuity equation

$$\partial_t\rho + \nabla\cdot(\rho\mathbf{v}_s) = 0, \qquad \mathbf{v}_s = \frac{\nabla S}{m}, \tag{10}$$

and the Hamilton-Jacobi equation with quantum-pressure correction

$$\partial_t S + \frac{|\nabla S|^2}{2m} + V + Q = 0, \qquad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}. \tag{11}$$

This is the Madelung form [madelung1927] of the Schrödinger equation. The continuity equation and the Hamilton-Jacobi-with-Q pair are mathematically equivalent to equation~(9).

5. The Born Rule

A measurement is a commitment event that occurs when environmental bandwidth grows rapidly enough to force an individuation boundary between the chain and its environment. At the commitment event, the chain transitions from a multi-channel coherent state to a single-channel state, and a new micro-event is added at the committed channel's next-vertex position.

5.1 Environmental phase-randomization

The environmental coupling that triggers a commitment event induces random phase shifts on the chain's channels:

$$P_K(x,t) \;\longmapsto\; P_K(x,t)\, e^{i\delta_K(x,t)}, \tag{12}$$

where δ_K are random phases drawn from the environmental-mode distribution. The phases $\{\delta_K\}$ are statistically independent across channels K. Independence is forced by three independent primitive-level arguments: the four-band decomposition of equation~(1) assigns the environmental bandwidth of each channel to a distinct subset of environmental modes; the locality of commitment events excludes long-range environmental correlations between channels; and the orthogonality of environmental modes, which are eigenmodes of a Hermitian bath Hamiltonian, renders the phases of distinct modes independent. Moreover, any correlation between δ_K and $\delta_{K'}$ for $K \neq K'$ would violate bandwidth conservation, the locality of commitment, and the no-signaling property of entangled bipartite systems; all three violations are structurally excluded.

5.2 Destruction of inter-channel coherence

Consider the squared magnitude of the coherent sum,

$$|\Psi|^2 = \sum_K |P_K|^2 + \sum_{K\neq K'} P_K^*\, P_{K'}. \tag{13}$$

Under the phase-kick of equation~(12), the cross-terms acquire random phase factors $e^{i(\delta_{K'}-\delta_K)}$. Averaging over the environmental phase distribution,

$$\langle e^{i(\delta_{K'}-\delta_K)}\rangle_{\text{env}} = \delta_{K,K'} \tag{14}$$

for independent phases, so the cross-terms vanish on environmental averaging. The surviving diagonal terms $\sum_K |P_K|^2$ constitute an incoherent mixture in the channel basis.

5.3 Selection probability

The incoherent mixture assigns to channel K a weight $|P_K|^2$. Normalizing this weight by the total bandwidth $\sum_K |P_K|^2$ yields the selection probability

$$\text{Prob}(K^*) = \frac{|P_{K^*}(x)|^2}{\sum_K |P_K(x)|^2}. \tag{15}$$

Equation~(15) is the Born rule [born1926]. Under the identification $\Psi = \sum_K P_K$, the amplitude in channel K equals the projection $\langle K|\Psi\rangle$, and equation~(15) reads $\text{Prob}(K^*) = |\langle K^*|\Psi\rangle|^2/\langle\Psi|\Psi\rangle$, recovering the conventional quantum-mechanical statement.

The squared-amplitude form of equation~(15) is definitional rather than postulated. The identification $b_K = |P_K|^2$ from equation~(3) makes the bandwidth-fraction and the squared-amplitude-fraction the same quantity. The Born rule's exponent of two is not an independent empirical fact requiring its own justification; it is a consequence of the structural choice that constructs the bandwidth from the participation measure by squaring the amplitude. Had the decomposition been $P_K = b_K e^{i\pi_K}$ with $b_K = |P_K|$ instead, the Born rule would read $\mathrm{Prob}(K^*) = |P_{K^*}|/\sum_K |P_K|$. The empirical consistency of the squared-amplitude form under Madelung anchoring forces equation~(3).

6. Bell Correlations and the Tsirelson Bound

Entanglement arises in the ED framework from shared participation structure across two or more endpoints of a bipartite or multipartite system. The joint participation measure of a bipartite system is a complex-valued function

$$P^{AB}_{K_A,K_B}(x_A, x_B, t) \in \mathbb{C} \tag{16}$$

over the product channel space. A joint state is factorizable if $P^{AB}_{K_A K_B} = P^A_{K_A} \cdot P^B_{K_B}$, and non-factorizable (entangled) otherwise.

6.1 The CHSH scenario

Consider a bipartite system with subsystems A and B spatially separated. Each subsystem admits two measurement settings, a and a' for A and b and b' for B, and each measurement produces a binary outcome ± 1. The correlation functions

$$E(a, b) = \sum_{s_A, s_B = \pm 1} s_A \, s_B \, \mathrm{Prob}(s_A, s_B \,|\, a, b) \tag{17}$$

combine into the CHSH statistic

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b'). \tag{18}$$

The Bell inequality [bell1964] states that $|S| \leq 2$ for any local hidden-variable theory; the CHSH formulation [chsh1969] is the specific form of equation~(18). The Tsirelson bound [tsirelson1980] states that $|S| \leq 2\sqrt{2}$ under standard quantum mechanics.

6.2 Factorizable joint participation

For factorizable joint participation, the joint probability factors: $\mathrm{Prob}(K_A, K_B \,|\, a, b) = \mathrm{Prob}_A(K_A \,|\, a) \cdot \mathrm{Prob}_B(K_B \,|\, b)$. Each subsystem's outcome depends on local measurements and local randomness only. Standard Bell-CHSH algebra gives

$$S = \langle s_A(a)[s_B(b) + s_B(b')]\rangle + \langle s_A(a')[s_B(b) - s_B(b')]\rangle. \tag{19}$$

For realizations of the local random variables with $s_B(\cdot) \in \{-1, +1\}$, one of $s_B(b) + s_B(b')$ and $s_B(b) - s_B(b')$ vanishes while the other has magnitude two. Consequently $|S| \leq 2$, which is the Bell-CHSH inequality.

6.3 Non-factorizable joint participation

For non-factorizable joint participation, the joint probability does not factor. Consider the prototypical singlet configuration

$$P^{AB}_{K_A K_B} = \frac{1}{\sqrt{2}}\big(\delta_{K_A,0}\delta_{K_B,1} - \delta_{K_A,1}\delta_{K_B,0}\big), \tag{20}$$

representing a pair of two-level subsystems in the spin-singlet state. With measurement settings corresponding to Pauli observables σ_a, σ_b on unit vectors $\mathbf{a}, \mathbf{b}$, the correlation function is

$$E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}. \tag{21}$$

Choosing angles $\mathbf{a} = \hat{\mathbf{x}}$, $\mathbf{a}' = \hat{\mathbf{z}}$, $\mathbf{b} = (\hat{\mathbf{x}} + \hat{\mathbf{z}})/\sqrt{2}$, $\mathbf{b}' = (\hat{\mathbf{x}} - \hat{\mathbf{z}})/\sqrt{2}$ yields $S = -2\sqrt{2}$, so $|S| = 2\sqrt{2}$, saturating the Tsirelson bound.

6.4 The Tsirelson bound from participation-manifold structure

Define the CHSH operator

$$\hat{S} = \sigma_a \otimes \sigma_b + \sigma_a \otimes \sigma_{b'} + \sigma_{a'} \otimes \sigma_b - \sigma_{a'} \otimes \sigma_{b'}. \tag{22}$$

Its square satisfies

$$\hat{S}^2 = 4\, I_A \otimes I_B + [\sigma_a, \sigma_{a'}] \otimes [\sigma_b, \sigma_{b'}], \tag{23}$$

using $\sigma_a^2 = I$ for Hermitian unit-norm observables. The operator-norm bound $\|[\sigma, \sigma']\| \leq 2$ for such observables gives $\|\hat{S}^2\| \leq 4 + 2 \cdot 2 = 8$, so $\|\hat{S}\| \leq 2\sqrt{2}$. Since $|S| = |\langle P^{AB}|\hat{S}|P^{AB}\rangle| \leq \|\hat{S}\| \cdot \|P^{AB}\|^2$ and the joint participation measure is normalized, $|S| \leq 2\sqrt{2}$. This Landau-Khalfin argument [landau1987] rests on the sesquilinear inner-product structure of equation~(6), the normalization of the joint participation measure, and the unit-norm Hermitian character of the measurement observables, all of which are primitive-level or U-derived properties of the participation-manifold.

The no-signaling property of the joint participation measure follows from the locality of commitment events. The marginal probability on A is independent of the measurement setting at B; this is structural rather than postulated.

7. The Heisenberg Uncertainty Relations

The uncertainty relations arise in the ED framework from the Fourier-conjugate decomposition of the adjacency-band bandwidth budget. The adjacency bandwidth b^{adj} encodes two structurally distinct kinds of nearness: spatial nearness, associated with localization in the position coordinate x, and phase-propagation nearness, associated with localization in the wavenumber k or momentum $p = \hbar k$. The two kinds of adjacency are connected by the Fourier transform of the coherent-sum wavefunction.

7.1 Adjacency-band partition

The adjacency band admits a two-component orthogonal decomposition

$$b^{\text{adj}} = b_x + b_p, \tag{24}$$

where b_x is the position-adjacency bandwidth, encoded in the spatial density distribution $|\Psi(x)|^2$, and b_p is the momentum-adjacency bandwidth, encoded in the momentum density distribution $|\tilde{\Psi}(p)|^2$. The two distributions are related by the Fourier transform

$$\tilde{\Psi}(p,t) = (2\pi\hbar)^{-1/2} \int dx\, \Psi(x,t)\, e^{-ipx/\hbar}. \tag{25}$$

Equation~(25) is the unique basis-change between position and momentum representations under translation invariance of the participation graph; Stone's theorem identifies $\hat{P} = -i\hbar\partial_x$ as the generator of translations, and the Fourier transform is the basis-change to the eigenbasis of $\hat{P}$. Parseval's theorem $\int dx|\Psi|^2 = \int dp|\tilde{\Psi}|^2$ shows that the two distributions are complementary decompositions of the same total adjacency bandwidth. The Stone-von Neumann theorem establishes that the Heisenberg-Weyl algebra generated by $\hat{x}$ and $\hat{P}$ is irreducible on $L^2(\mathbb{R})$, so no third independent decomposition component exists in one spatial dimension.

7.2 Derivation of the uncertainty relation

The bandwidth-allocation inequality for the orthogonal partition~(24) takes the form

$$(\Delta b_x)(\Delta b_p) \geq K_{xp}, \tag{26}$$

where K_{xp} is a structural constant. Under the identifications $b_x(x) \propto |\Psi(x)|^2$ and $b_p(p) \propto |\tilde{\Psi}(p)|^2$ inherited from the Born-rule structure applied within each conjugate partition, the bandwidth spreads are the standard position and momentum variances of the wavefunction,

$$(\Delta b_x)^2 = \langle x^2 \rangle_{|\Psi|^2} - \langle x \rangle^2_{|\Psi|^2} = (\Delta x)^2, \tag{27}$$

and similarly $(\Delta b_p)^2 = (\Delta p)^2$. The mathematical Fourier-uncertainty theorem applied to the Fourier pair $(\Psi, \tilde{\Psi})$ with the canonical commutator $[\hat{x}, \hat{P}] = i\hbar$ yields [heisenberg1927, robertson1929]

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}. \tag{28}$$

The bound saturates for Gaussian minimum-uncertainty wavepackets. Equation~(28) is the Heisenberg position-momentum uncertainty relation. Unitary evolution under equation~(9) preserves the inequality at all times.

7.3 Generalizations

The four-band structure supports analogous decompositions for other conjugate pairs. The internal rule-bandwidth and the commitment-reserve bandwidth form a conjugate pair for which the energy-time uncertainty relation holds. The commitment-reserve bandwidth and the rule-phase form a conjugate pair for which the number-phase uncertainty relation holds. Directional decompositions of the internal rule-bandwidth support the angular-momentum uncertainty relations. In each case, the same allocation-inequality mechanism within a specific band-pair of the four-band structure produces the corresponding uncertainty relation. The universal appearance of $\hbar/2$ as the bound reflects the common Fourier-conjugacy content, inherited through the anchoring of $\hbar$ at the dimensional level.

8. The Exponent-Two Thread

The five-step derivation of sections 3 through 7 produces the quantum-mechanical content through a single structural commitment that threads through otherwise-independent appearances of the number two.

8.1 Five appearances

The Born rule selection probability $\mathrm{Prob}(K^*) = |P_{K^*}|^2 / \sum_K |P_K|^2$ contains the squared amplitude. The Schrödinger kinetic term $\hat{H}_{\mathrm{kin}} = \hbar^2 k^2/(2m)$ is quadratic in the momentum. The Madelung decomposition $\Psi = \sqrt{\rho}\, e^{iS/\hbar}$ contains the square root of the density. The sublinear bandwidth-composition rule expresses the combined bandwidth of two channels in terms of squared-amplitude-level cross-terms $b^2_{\mathrm{combined}} = b_1^2 + b_2^2 + 2c_{12}b_1b_2$. The Fourier-uncertainty bound $(\Delta x)^2(\Delta p)^2 \geq (\hbar/2)^2$ is quadratic in the variances. Each of these five appearances involves the exponent two in a way that is not, in conventional quantum mechanics, tied to a single underlying structural commitment.

8.2 The single underlying commitment

In the participation-measure framework, each of the five appearances arises from the same definitional commitment: bandwidth is the squared amplitude of the participation measure, $b_K = |P_K|^2$ (equation~(3)). The Born rule's squared amplitude is the bandwidth fraction by construction. The Schrödinger kinetic term's $\hbar^2 k^2/(2m)$ is the quadratic leading-order dispersion forced by translation and rotation invariance; the $\hbar^2$ factor arises from the square of the momentum operator acting on the wavefunction, which is the coherent sum of participation amplitudes. The Madelung decomposition's $\sqrt{\rho}$ is the amplitude $|\Psi|$ whose square is the density. The sublinear composition rule's squared-amplitude cross-structure is the composition law for the squared magnitudes of the participation measure. The Fourier-uncertainty bound's squared variances arise from the second-moment structure of the conjugate-partition densities. All five are manifestations of the same structural choice.

In conventional quantum mechanics, these five appearances are logically independent. The Born exponent could in principle differ from the kinetic exponent without immediate empirical inconsistency; Gleason's theorem and related arguments fix the Born exponent from Hilbert-space axioms, but do not tie it to the kinetic exponent. In the participation-measure framework, the five exponents are tied together structurally: a change in the definitional identification $b = |P|^2$ would change all five. The structural unification is the program's strongest positive claim.

9. Structural Content versus Dimensional Anchoring

The derivations of sections 3 through 7 produce the form of the quantum-mechanical equations — the Schrödinger structure, the Born probability formula, the Bell-CHSH inequality and Tsirelson bound, and the Heisenberg uncertainty relation — from the primitive-level participation-measure framework. The numerical values of the dimensional constants that enter these equations are not derived from primitives. The constant $\hbar$ is anchored through the ED Dimensional Atlas via the Madelung identification of the quantum regime's diffusion coefficient $D_{\mathrm{phys}} = \hbar/(2m)$ with the free-particle Schrödinger equation; $\hbar$ enters equation~(7) with its empirically measured value. The mass parameter m is anchored empirically at the species level; the primitive-level derivation of particle masses from chain structure remains an open research problem and is flagged in section 10.

The separation is clean at the level of the derivation. All structural content — linearity of the evolution equation, first-order-in-time structure, operator-form decomposition of the evolution kernel, norm preservation and the associated anti-Hermitian generator, complex-valued-ness of the wavefunction, sesquilinear inner-product structure, Fourier-conjugate partition of the adjacency band, the squared-amplitude form of the Born rule, the Bell-CHSH factorizable bound, the Tsirelson bound, the Heisenberg inequality — is derived from primitives combined with standard mathematical theorems (Stone's theorem, Parseval's theorem, Stone-von Neumann theorem, Fourier-uncertainty

theorem, Landau-Khalfin operator-norm algebra). The dimensional constants $\hbar$ and m are inherited via the Dimensional Atlas from empirical measurement. No other content is inherited.

The framework does not predict the numerical value of $\hbar$; it inherits it. The framework does not predict the mass spectrum of elementary particles; these are inherited at the species level and constitute open research questions. The framework does not make empirical predictions that differ from standard quantum mechanics within the tested thin-participation regime; retrodiction attempts against matter-wave interferometry data in the Eibenberger and Fein regimes reproduce the standard predictions within the consistent-weak-form epistemic category.

10. Limitations and Research Frontiers

The structural reduction presented in this paper has explicit limitations and leaves several research problems open for future work.

First, the primitive-level derivation of particle masses from chain structure is flagged as speculative. The chain-mass problem asks how the specific rule-type bandwidth signature of an electron, proton, or other elementary particle is set by the underlying participation-graph structure. A first-pass candidate identification of the rest mass with the rule-type bandwidth divided by c^2 has been articulated, but a rigorous derivation requires a primitive-level derivation of the rule-type taxonomy, which is itself an open question. The problem connects to the origin of the Standard-Model mass spectrum and is a multi-year research frontier.

Second, the numerical value of $\hbar$ is inherited rather than derived. The Dimensional Atlas anchors $\hbar$ through the Madelung-class identification of the quantum regime; a primitive-level origin for the specific numerical value of $\hbar$ would reduce the inheritance further but has not been attempted.

Third, the derivation is non-relativistic. The relativistic extension would replace the quadratic dispersion $H_k = \hbar^2 k^2/(2m)$ with the relativistic dispersion $E = \sqrt{p^2c^2 + m^2c^4}$, requiring a Lorentz-covariant formulation of the participation measure and a corresponding covariant four-band decomposition. The Klein-Gordon and Dirac equations would arise as the thin-limit consequences of the covariant framework; neither has been derived in the present work.

Fourth, the derivation is single-particle. A multi-particle or quantum-field-theoretic extension would require the development of second-quantized participation measures, with particle-creation and particle-annihilation operators emerging as primitive-level structural consequences. The connection to the Standard Model field content would require the rule-type taxonomy derivation mentioned above.

Fifth, the evolution equation~(7) has been derived in the isolated-chain thin-participation limit, where environmental coupling and commitment events are suppressed. The extension to non-isolated chains would produce a Lindblad-master-equation-type structure with jump operators corresponding to environmental commitment events; the primitive-level derivation of this extended dynamics is near-term future work.

Sixth, the framework does not currently make empirical predictions that distinguish it from standard quantum mechanics in the regime where standard quantum mechanics has been tested. Distinguishing content is expected to emerge in regimes where the thin-participation limit breaks down, including the quantum-to-classical transition at macroscopic scale and cosmological-scale dynamics relevant to baryogenesis; preliminary retrodiction attempts in these regimes have been structurally informative but not empirically distinguishing within currently available data.

11. Conclusion

The non-relativistic quantum mechanics of a single particle has been shown to arise as a structural consequence of the Event Density primitive stack. A single mathematical object — the participation measure $P_K(x,t) = \sqrt{b_K(x,t)}\, e^{i\pi(K,x,t)}$ — serves as the carrier of quantum-mechanical content. Its thin-participation limit produces the Schrödinger equation; commitment events under environmental phase-randomization produce the Born rule in its squared-amplitude form; non-factorizable joint participation measures produce Bell-inequality-violating correlations saturating at the Tsirelson bound; Fourier-conjugate partition of the adjacency bandwidth produces the Heisenberg uncertainty relations. The five otherwise-independent appearances of the exponent two in quantum mechanics — the Born rule, the Schrödinger kinetic term, the Madelung decomposition, the sublinear bandwidth-composition rule, and the Fourier-uncertainty bound — are manifestations of a single definitional commitment, the identification of bandwidth with squared amplitude. No additional physical postulate beyond the participation-measure structure, the four-band bandwidth decomposition, and the commitment dynamics is required. The numerical values of $\hbar$ and m are inherited through the ED Dimensional Atlas; the primitive-level derivation of particle masses remains an open research problem. The result is a structural reduction of the four independent postulates of standard quantum mechanics — state structure, unitary evolution, measurement probability, and observable-commutator uncertainty — to structural consequences of a single primitive-level framework.

References

[bell1964] Bell, J. S. (1964). On the Einstein-Podolsky-Rosen paradox. *Physics Physique Fizika*, **1**, 195–200.

[bohm1952] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables. I and II. *Physical Review*, **85**, 166–179, 180–193.

[born1926] Born, M. (1926). Zur Quantenmechanik der Stoßvorgänge. *Zeitschrift für Physik*, **37**, 863–867.

[chsh1969] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, **23**, 880–884.

[fuchs2002] Fuchs, C. A. (2002). Quantum mechanics as quantum information (and only a little more). *arXiv:quant-ph/0205039*.

[gleason1957] Gleason, A. M. (1957). Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, **6**, 885–893.

[heisenberg1927] Heisenberg, W. (1927). Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Zeitschrift für Physik*, **43**, 172–198.

[landau1987] Landau, L. J. (1987). On the violation of Bell's inequality in quantum theory. *Physics Letters A*, **120**, 54–56.

[madelung1927] Madelung, E. (1927). Quantentheorie in hydrodynamischer Form. *Zeitschrift für Physik*, **40**, 322–326.

[robertson1929] Robertson, H. P. (1929). The uncertainty principle. *Physical Review*, **34**, 163–164.

[rovelli1996] Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics*, **35**, 1637–1678.

[stone1930] Stone, M. H. (1930). Linear transformations in Hilbert space, III: operational methods and group theory. *Proceedings of the National Academy of Sciences*, **16**, 172–175.

[tsirelson1980] Tsirelson, B. S. (1980). Quantum generalizations of Bell's inequality. *Letters in Mathematical Physics*, **4**, 93–100.

[wallace2012] Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation*. Oxford University Press.